WIRELESS COMMUNICATION – 2 MARKS

1. Differentiate Linear and Non-linear Equalization

Linear: Output is a linear function of the input. Examples: Zero Forcing and LMS equalizers.

Non-linear: Output is a non-linear function of the input. Example: Decision Feedback Equalizer (DFE).

2. Explain the effect of diversity on error probability in fading channels.

Diversity provides multiple independently faded versions of the same signal. It reduces deep-fading effects and hence reduces the probability of error (BER). Greater diversity gives better performance.

3. Write the MIMO system model/equation.

Y = HX + N

Y → received signal vector

H → channel matrix

X → transmitted signal vector

N → noise vector

4. Define pre-coding and beam forming.

Pre-coding: Processing the transmitted signals before transmission to improve signal quality and reduce interference.

Beamforming: Adjusting the phase and amplitude of signals from multiple antennas to direct the signal toward the desired receiver.

5. What is ISI? How does equalization overcome it?

ISI (Inter-Symbol Interference): Interference of one transmitted symbol with neighboring symbols due to multipath propagation and channel distortion.

Equalization compensates for channel distortion and reduces ISI, helping to recover the transmitted symbols correctly.

DIFFERENCE BETWEEN LINEAR AND NON-LINEAR EQUALIZATION

Equalization is the process of reducing Inter Symbol Interference (ISI) caused by a channel so that the received signal can be detected accurately. Equalizers are mainly classified into Linear Equalizers and Non-linear Equalizers.

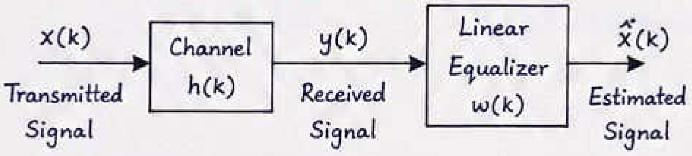
1. Linear Equalization :

Linear equalizers use a linear system (addition and multiplication only) to approximate the inverse of the channel. They are simple in structure and easy to implement.

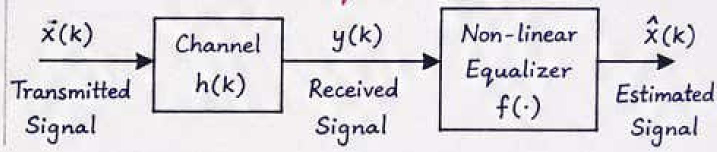
2. Non-linear Equalization :

Non-linear equalizers use non-linear operations to compensate for severe distortions in the channel. They can handle large ISI and are more effective in challenging channels.

Linear Equalizer Model



Non-linear Equalizer Model



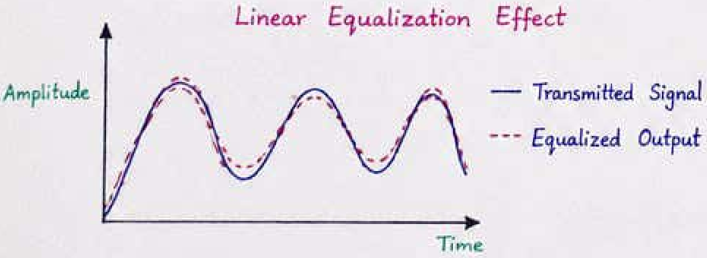
3. Key Differences :

Parameter	Linear Equalization	Non-linear Equalization
1. Operation	Uses linear operations (addition and multiplication only).	Uses non-linear operations (e.g., decision, lookup table, clipping, etc.).
2. Structure	Simple and easy to implement (FIR or IIR filters).	Complex structure (decision feedback, lookup tables, neural networks, etc.).
3. ISI Compensation	Effective for small to moderate ISI.	Effective even for severe ISI.
4. Performance	Lower performance in highly non-linear or severe channels.	Higher performance in non-linear and time-varying channels.
5. Complexity	Low computational complexity.	High computationd complexity.
6. Adaptation	Easily adapted using algorithms like LMS, RLS.	Adaptation is difficult; may require training or iterative methods.
7. Examples	Zero Forcing Equalizer, MMSE Equalizer, LMS Equalizer.	Decision Feedback Equalizer (DFE), Maximum Likelihood Sequence Estimator, Neural Equalizer.
8. Application	Used in systems with mild ISI and linear channels.	Used in systems with severe ISI, non-linear distortion and fading channels.

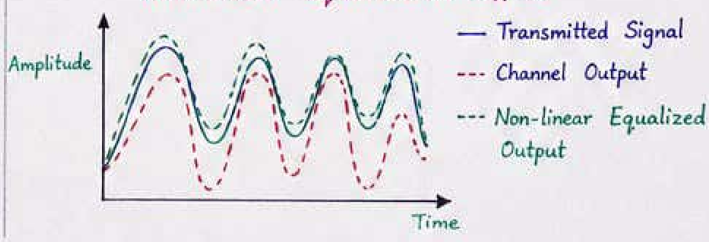
4. Summary :

Linear equalizers are simple and suitable for channels with mild distortion, while non-linear equalizers are complex but provide better performance in channels with severe ISI and non-linear effects.

Linear Equalization Effect



Non-linear Equalization Effect



★ Thus, the choice between linear and non-linear equalization depends on the channel condition, required performance and complexity constraints. ★

DIVERSITY COMBINING TECHNIQUES

Diversity Combining is a technique used in wireless communication systems to improve the reliability of the received signal. It uses multiple independently faded copies of the same signal and combines them at the receiver to reduce the effect of fading. The combined signal has higher Signal-to-Noise Ratio (SNR) and hence lower probability of error.

Types of Diversity Combining :

1. Selection Combining (SC)
2. Equal Gain Combining (EGC)
3. Maximal Ratio Combining (MRC)

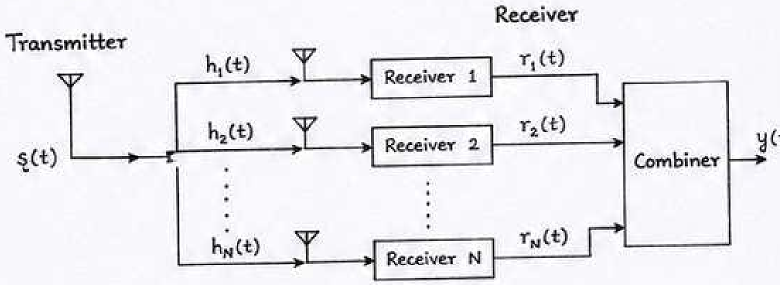


Fig: General diversity combining system

1. SELECTION COMBINING (SC):

- In selection Combining, the receiver output with the highest instantaneous SNR is selected and used as the output.
- It is the simplest diversity technique and does not require amplitude or phase information.
- It provides improvement in performance compared to single receiver but less than other techniques.

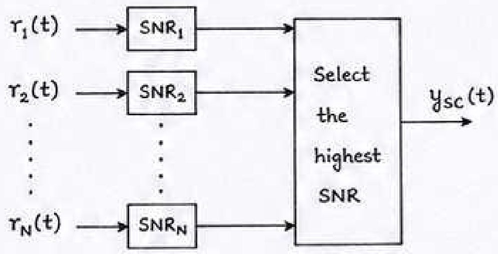


Fig: Selection Combining

Features:
- Simple
- No phase info needed
- Lower complexity
- Less performance improvement

2. EQUAL GAIN COMBINING (EGC):

- In EGC, all received signals are co-phased with respect to a reference signal and then added with equal weights.
- Only phase adjustment is required, amplitudes are not changed.
- Provides better performance than SC but less than MRC.

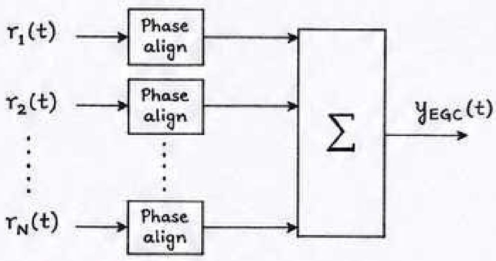


Fig: Equal Gain Combining

Features:
- Requires phase information
- Equal weights
- Moderate complexity
- Better than SC

3. MAXIMAL RATIO COMBINING (MRC):

- In MRC, each received signal is multiplied by a weight proportional to its SNR (or channel gain) and then all are added.
- It maximizes the output SNR and minimizes the probability of error.
- Provides the best performance among the three techniques.

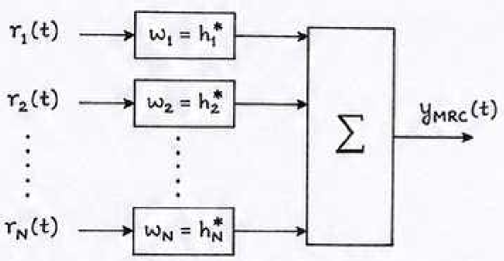


Fig: Maximal Ratio Combining

Features:
- Requires both amplitude and phase information
- Weights proportional to SNR
- Highest complexity
- Best performance

COMPARISON :

Technique	Complexity	Requirement	Performance (SNR)
Selection Combining	Low	No phase info	Lowest
Equal Gain Combining	Medium	Phase info	Better than SC
Maximal Ratio Combining	High	Amplitude & Phase info	Best

Advantages of Diversity Combining :
- Reduces fading effects.
- Improves reliability.
- Decreases probability of error.
- Extends coverage area.

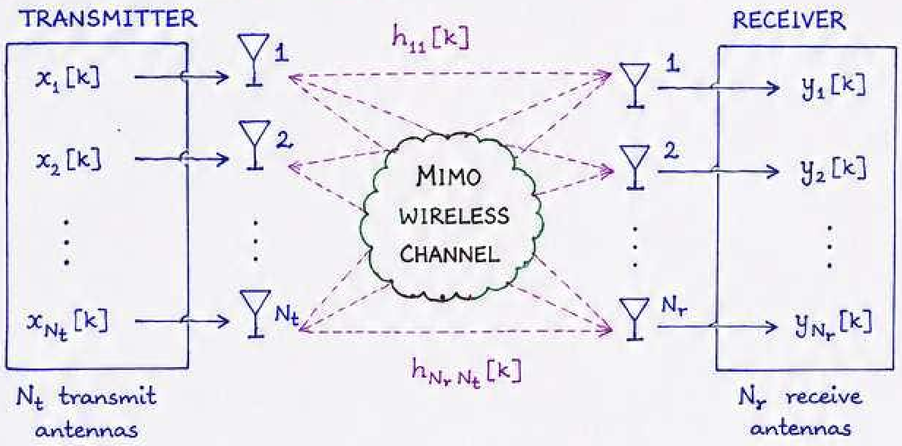
Applications :
- Used in mobile communication, satellite systems, Wi-Fi, 4G/5G, etc.

MIMO SYSTEMS – DEFINITION AND BASIC INPUT-OUTPUT SYSTEM MODELS

1. DEFINITION :

MIMO (Multiple-Input Multiple-Output) system is a wireless communication system that employs multiple transmit antennas at the transmitter and multiple receive antennas at the receiver. It exploits multipath propagation to send and receive independent data streams simultaneously over the same frequency band, thereby improving channel capacity, reliability and spectral efficiency.

2. BASIC MIMO SYSTEM MODEL :



where,

- N_t : Number of transmit antennas
- N_r : Number of receive antennas
- $x_i[k]$: Signal transmitted from i^{th} transmit antenna
- $y_j[k]$: Signal received at j^{th} receive antenna
- $h_{ji}[k]$: Channel coefficient from i^{th} transmit antenna to j^{th} receive antenna
- $n_j[k]$: Additive noise at j^{th} receive antenna

3. INPUT-OUTPUT RELATIONSHIP :

(a) Discrete-Time Baseband Model :

The received signal at the j^{th} receive antenna is

$$\boxed{y_j[k] = \sum_{i=1}^{N_t} h_{ji}[k]\, x_i[k] + n_j[k] \quad , \quad j = 1, 2, \ldots, N_r} \longrightarrow (1)$$

In vector-matrix form,

$$\boxed{\mathbf{y}[k] = \mathbf{H}[k]\ \mathbf{x}[k] + \mathbf{n}[k]} \longrightarrow (2)$$

where,

$$\mathbf{x}[k] = [x_1[k] \ \ x_2[k] \ \cdots \ x_{N_t}[k]]^T \quad (N_t \times 1)$$

$$\mathbf{y}[k] = [y_1[k] \ \ y_2[k] \ \cdots \ y_{N_r}[k]]^T \quad (N_r \times 1)$$

$$\mathbf{n}[k] = [n_1[k] \ \ n_2[k] \ \cdots \ n_{N_r}[k]]^T \quad (N_r \times 1)$$

$$\mathbf{H}[k] = \begin{bmatrix} h_{11}[k] & h_{12}[k] & \cdots & h_{1N_t}[k] \\ h_{21}[k] & h_{22}[k] & \cdots & h_{2N_t}[k] \\ \vdots & \vdots & \ddots & \vdots \\ h_{N_r1}[k] & h_{N_r2}[k] & \cdots & h_{N_rN_t}[k] \end{bmatrix} \quad (N_r \times N_t)$$

> Each element $h_{ji}[k]$ represents the subchannel between transmit antenna i and receive antenna j at time instant k.

(b) Continuous-Time Baseband Model :

The input-output relation in continuous time is

$$\boxed{y_j(t) = \sum_{i=1}^{N_t} h_{ji}(t) * x_i(t) + n_j(t) \quad , \quad j = 1, 2, \ldots, N_r} \longrightarrow (3)$$

In vector form,

$$\boxed{\mathbf{y}(t) = \mathbf{H}(t) * \mathbf{x}(t) + \mathbf{n}(t)} \longrightarrow (4)$$

where, $*$ denotes convolution operation.

4. REMARKS :

- If $N_t = 1$ or $N_r = 1$, the system reduces to SISO (Single Input Single Output).
- When $N_t, N_r > 1$, spatial multiplexing gain (multiple data streams) is achieved.
- The matrix $\mathbf{H}$ contains the complete channel state information (CSI).

> Hence, the basic input-output model of a MIMO system is represented by
> $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$ (discrete time) or $\mathbf{y}(t) = \mathbf{H}(t) * \mathbf{x}(t) + \mathbf{n}(t)$ (continuous time).

BASIC PRINCIPLES OF ZERO FORCING AND LMS EQUALIZATION ALGORITHMS

1. ZERO FORCING (ZF) EQUALIZATION

Zero Forcing equalization aims to completely eliminate ISI by forcing the overall channel response to be a delayed impulse.

Basic Principle :

- Let $H(z)$ be the channel transfer function.
- ZF equalizer has transfer function $G_{ZF}(z)$.
- The overall system response :

$$H(z)\, G_{ZF}(z) = k z^{-d}$$

  where k is a constant gain and d is the delay.
- This means the combined response is ideally an impulse at delay $d \rightarrow$ no ISI.

Derivation :

- ZF equalizer tries to invert the channel.

$$\therefore \quad G_{ZF}(z) = \frac{1}{H(z)}$$

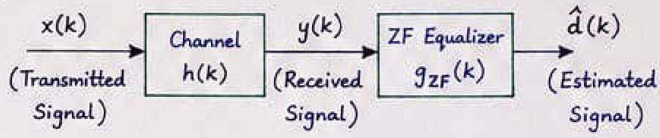
- In practice, to avoid noise enhancement, a finite length approximation is used.
- For frequency response :

$$G_{ZF}(e^{j\omega}) = \frac{1}{H(e^{j\omega})}$$

- Output : $\hat{d}(k) = (x * h * g_{ZF})(k) = k\, d(k-d) + n'(k)$

  where $n'(k)$ is enhanced noise.

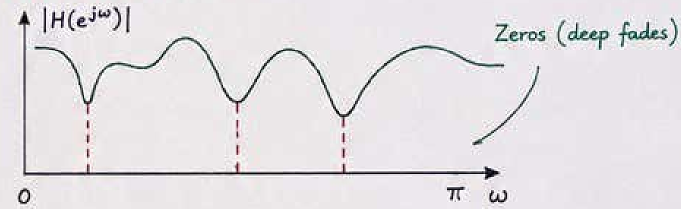
ZF Equalization Structure



Features :

- Completely removes ISI (in theory).
- Very sensitive to noise (especially when $H(e^{j\omega})$ has deep fades or zeros).
- Requires accurate channel knowledge.
- Often implemented as FIR filter.

Notes / Illustration :



At these frequencies
$H(e^{j\omega}) \approx 0 \rightarrow G_{ZF}(e^{j\omega})$ becomes very large $\rightarrow$ noise enhancement.

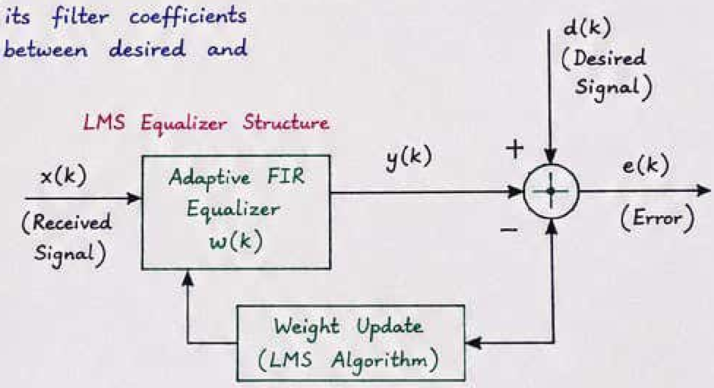
2. LMS (LEAST MEAN SQUARE) EQUALIZATION

LMS equalizer is an adaptive equalizer that adjusts its filter coefficients iteratively to minimize the mean square error (MSE) between desired and actual output.

Basic Principle :

- The equalizer is modeled as an FIR filter with adjustable coefficients.
- The output is compared with a desired (known or training) signal.
- Error is used to update the coefficients to minimize MSE.

LMS Equalizer Structure



Algorithm Steps :

1. Initialize weight vector $w(0)$ (usually zeros).
2. For each time instant k :
   - Compute equalizer output : $y(k) = w^T(k)\, x(k)$
   - Compute error : $e(k) = d(k) - y(k)$
   - Update weights :

$$\boxed{w(k+1) = w(k) + \mu\, e^*(k)\, x(k)}$$

where μ = step size $\left(0 < \mu < \frac{2}{\lambda_{max}}\right)$

(λ_{max} = maximum eigenvalue of R_{xx})

Features :

- Does not require exact channel knowledge.
- Adapts to time-varying channels.
- Simple and easy to implement.
- Performance depends on step size μ.
- Converges to Wiener solution.

Notes :

- Small $\mu \rightarrow$ slow convergence but stable.
- Large $\mu \rightarrow$ fast convergence but may become unstable.
- LMS minimizes $J = E[|e(k)|^2]$ (mean square error).

Comparison (ZF vs LMS)

Parameter	ZF Equalization	LMS Equalization
Approach	Non-adaptive (Inversion)	Adaptive (Iterative)
Channel Knowledge	Requires exact $H(z)$	Not required
ISI Elimination	Complete (in theory)	Approximate (depends on convergence)
Noise Sensitivity	High	Low to Moderate
Complexity	Higher (due to inversion)	Low
Best Use	Static channels	Time-varying channels

Summary :

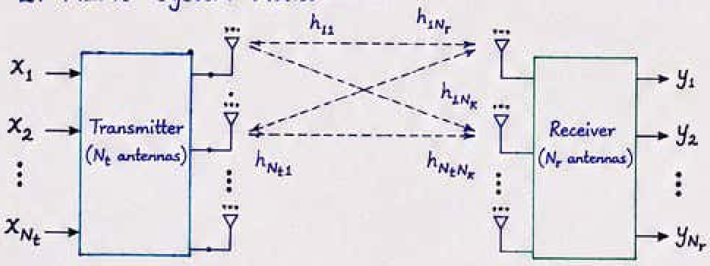
ZF equalizer inverts the channel to cancel ISI but amplifies noise at deep fades.
LMS equalizer adapts its coefficients to minimize the error and is suitable for unknown or time-varying channels. ★

1) Channel State Information (CSI) and Its Significance in MIMO Systems

1. Definition:

Channel State Information (CSI) refers to the knowledge of the properties of the wireless channel between the transmitter and receiver. In MIMO systems, it is usually represented by the channel matrix H, which contains the complex gains (magnitude and phase) of all propagation paths between transmit and receive antennas.

2. MIMO System Model:



$$\boxed{y = Hx + n}$$

where
- y : Received signal vector ($N_r \times 1$)
- x : Transmitted signal vector ($N_t \times 1$)
- H : Channel matrix ($N_r \times N_t$)
- n : Noise vector ($N_r \times 1$)

3. Acquisition of CSI:

- Obtained using pilot symbols (training sequences).
- Can be estimated at the receiver, the transmitter, or both.
- CSI at transmitter → requires feedback from receiver.

4. Significance of CSI in MIMO Systems:

① Optimal Detection:
CSI at receiver allows techniques like Maximum Likelihood (ML) or Zero-Forcing (ZF) detection to recover transmitted symbols with low error probability.

② Spatial Multiplexing (Capacity Gain):
Requires CSI at both transmitter and receiver to achieve full multiplexing gain by independent data streams.

③ Beamforming (Array Gain):
CSI at transmitter enables beamforming to direct energy towards the receiver, improving SNR.

④ Interference Suppression:
Helps in separating desired signals from co-channel interference and multi-user interference.

⑤ Adaptive Modulation and Coding:
CSI helps in adapting transmission parameters (modulation order, coding rate, power) according to channel quality.

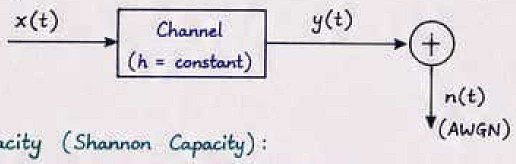
⑥ Reliability and Link Adaptation:
Provides information about fades and variations, enabling reliable communication through link adaptation.

Summary: CSI is the key enabler for realizing the full potential of MIMO systems. Without CSI, only limited diversity gain can be achieved. With perfect CSI at both ends, MIMO systems can achieve maximum capacity and reliability.

2) Difference Between the Capacity of Fading and Non-Fading Channels

A. Non-Fading Channel

- The channel gain is constant (h = constant).
- No variation with time, frequency or position.
- Only Additive White Gaussian Noise (AWGN) is present.



Capacity (Shannon Capacity):

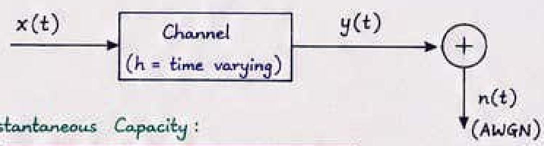
$$\boxed{C_{non\text{-}fading} = B \log_2\left(1 + \frac{P|h|^2}{N_0 B}\right)} \quad \text{bits/sec}$$

where,
- B : Bandwidth (Hz)
- P : Transmit power (Watts)
- N_0 : Noise power spectral density (W/Hz)
- h : Channel gain (constant)

Features:
- Capacity is constant for given P, B and h.
- Increases logarithmically with SNR ($= \frac{P|h|^2}{N_0 B}$).
- No uncertainty in channel.

B. Fading Channel

- The channel gain $h(t)$ varies randomly with time/frequency.
- Caused by multipath propagation, shadowing, etc.
- Leads to time-varying SNR.



Instantaneous Capacity:

$$\boxed{C_{fading}(t) = B \log_2\left(1 + \frac{P|h(t)|^2}{N_0 B}\right)} \quad \text{bits/sec}$$

Ergodic (Average) Capacity:

$$\overline{C}_{fading} = E_h\left[B \log_2\left(1 + \frac{P|h|^2}{N_0 B}\right)\right]$$

where $E_h[\cdot]$ = Expected value over fading distribution of $|h|^2$

Features:
- Capacity is random (changes with channel realizations).
- Average capacity is always less than non-fading capacity for the same average SNR.
- Deep fades cause capacity to drop close to zero.

C. Comparison Table

Parameter	Non-Fading Channel	Fading Channel
Channel Gain (h)	Constant	Time-varying (random)
SNR	Constant	Random (varies with time)
Capacity Expression	$C = B \log_2\left(1 + \frac{P\|h\|^2}{N_0 B}\right)$	$C(t) = B \log_2\left(1 + \frac{P\|h(t)\|^2}{N_0 B}\right)$
Capacity Nature	Deterministic (fixed value)	Random (instantaneous), Average capacity is considered
Effect of Fading	No fading → no degradation	Fading causes degradation in capacity
Reliability	More reliable (no deep fades)	Less reliable (deep fades possible)
Achievable Rate	Higher for same SNR	Lower for same average SNR

Key Point:
Fading channels introduce uncertainty and reduce the achievable capacity compared to non-fading channels. Techniques like diversity, adaptive modulation, and coding are used to combat this effect.